Lab #8: Paging Simulation

Page table:

```
Page Table (from entry 0 down to the max size)
  [       0]  0x80000018
  [       1]  0x00000000
  [       2]  0x00000000
  [       3]  0x8000000c
  [       4]  0x80000009
  [       5]  0x00000000
  [       6]  0x8000001d
  [       7]  0x80000013
  [       8]  0x00000000
  [       9]  0x8000001f
  [      10]  0x8000001c
  [      11]  0x00000000
  [      12]  0x8000000f
  [      13]  0x00000000
  [      14]  0x00000000
  [      15]  0x80000008

Virtual Address Trace
  VA 0x00003385 (decimal:    13189) --> PA or invalid address?
  VA 0x0000231d (decimal:     8989) --> PA or invalid address?
  VA 0x000000e6 (decimal:      230) --> PA or invalid address?
  VA 0x00002e0f (decimal:    11791) --> PA or invalid address?
  VA 0x00001986 (decimal:     6534) --> PA or invalid address?
```

Q1: How does page table size change?

Part A: Effect of Address Space Size

Given:

- Page Size = 1 KB

Calculations:

- 1 MB / 1 KB = 1024 entries
- 2 MB / 1 KB = 2048 entries
- 4 MB / 1 KB = 4096 entries

As the address space increases, the number of virtual pages increases, and thus the page table size grows linearly.

Part B: Effect of Page Size

Given:

- Address Space = 1 MB

Calculations:

- 1 MB / 1 KB = 1024 entries
- 1 MB / 2 KB = 512 entries
- 1 MB / 4 KB = 256 entries

As the page size increases, the number of pages decreases, so the page table becomes smaller.

Q2: Effect of Percentage of Allocated Pages

Given:

- Address Space = 16 KB
- Page Size = 1 KB → total = 16 pages
- u = 0 → no pages mapped → all accesses cause page faults
- u = 100 → all pages mapped → no page faults
- As u increases → number of valid mappings increases

Increasing the percentage of allocated pages reduces page faults and increases successful translations.

Q3: Unrealistic Parameter Combinations

1. -P 8 -a 32
   - Page size = 8 bytes, address space = 32 bytes
   - Unrealistic: extremely small and not used in real systems
2. -P 8k -a 32k
   - Only 4 pages
   - Valid but too small to be practical
3. -P 1m -a 256m
   - Large page size
   - Possible in some systems but may cause fragmentation

Very small page sizes or extremely small address spaces are unrealistic because they do not match real hardware constraints.

Q4: Virtual Address Translation

Page Size = 1 KB = 1024 bytes

Address Translations:

1. VA = 13189
   - Page Number = 12
   - Offset = 901
   - PTE[12] = valid (frame = 15)
   - Physical Address = 15 × 1024 + 901 = **16261**

---

2. VA = 8989
   - Page Number = 8
   - PTE[8] = invalid
   - **Segmentation Fault**

---

3. VA = 230
   - Page Number = 0
   - Offset = 230
   - PTE[0] = valid (frame = 24)
   - Physical Address = 24 × 1024 + 230 = **24806**

---

4. VA = 11791
   - Page Number = 11
   - PTE[11] = invalid
   - **Segmentation Fault**

---

5. VA = 6534
   - Page Number = 6
   - Offset = 390
   - PTE[6] = valid (frame = 29)
   - Physical Address = 29 × 1024 + 390 = **30086**

Final Answers Summary

- 13189 → 16261
- 8989 → Segmentation Fault
- 230 → 24806
- 11791 → Segmentation Fault
- 6534 → 30086
